

# Tony Lam

+1 (540) 815-9070 | [lam.tony540@gmail.com](mailto:lam.tony540@gmail.com) | [linkedin.com/in/lamtonycs](https://linkedin.com/in/lamtonycs) | [github.com/tonylam0](https://github.com/tonylam0)

## EDUCATION

### University of Virginia

Bachelor of Arts in Computer Science

Charlottesville, VA

August 2025 – May 2029

- **Relevant Coursework:** Data Structures & Algorithms, Software Development Essentials, Discrete Mathematics

## EXPERIENCE

### Director of Technology

May 2026 – Present

Bluelearn (Open-Source Nonprofit)

Remote

- Own technical direction for a **5,000+ member** open-source community with **90+ GitHub stars**, leading **20+ contributors** across a **pnpm monorepo** (React 19 web app, Hono API, shared schema package, database) that has shipped **1,000+ commits** through **100+ reviewed pull requests**.
- Architected a graph-based learning platform on **Cloudflare Workers** and **Supabase/PostgreSQL**, modeling subjects, objectives, and prerequisites as a traversable dependency graph and authoring the **Postgres functions** behind graph traversal, content versioning, and a role-based peer-review workflow with **Row-Level Security** enforcing access on every table.
- Established the contributor quality bar: a **150+ case integration test suite** running against live Postgres, **GitHub Actions CI**, and **Prettier/ESLint** pre-commit hooks, enabling first-time contributors to merge without breaking production.

### Software Engineering Intern

March 2026 – Present

CargoLabs

Charlottesville, VA

- Architected an isolated **dev deployment pipeline** on a dedicated **AWS** account, seeding **9 business domains** of mock data, so the team could test backend routes end-to-end without touching **production**.
- Created a new **serverless pricing API** for VIN-based trip-excess insurance, adding request validation and end-to-end pricing logic to launch a **new insurance product line**.
- Built **buy- and sell-side client search** (name, email, load/BOL) with server-side date filtering over new **Node.js / MySQL** endpoints, plus a **quote flow** with prior-answer autofill across the full stack.

## PROJECTS

### WallHax (Best AI & Data Science Hack) | Swift, ARKit, Python, React, Three.js, Luma AI

March 2026

- Developed a **real-time, multi-device AR** collaboration iOS app and web dashboard for shared situational awareness, winning **Best AI & Data Science Hack** at HooHacks against **850+ participants**.
- Engineered a **low-latency UDP heuristic engine** to translate live **ARKit SLAM** and IMU data into accurate 3D character states (stance, distance, head tracking) for remote peers.
- Leveraged **Luma AI** to process live mission frames into **photorealistic 3D Gaussian Splats**, rendering the post-mission digital environment with a frame-by-frame mission timeline via a custom **React** and **Three.js** web dashboard.

### Evergreen | React, Django, REST APIs, SQLite

November 2025 – February 2026

- Built a **full-stack platform** for sharing and discussing video essays on a **decoupled architecture** with a **Django REST** backend handling token authentication, content management, and threaded comments.
- Designed the **relational schema** for users, media, and nested comment threads, and built a responsive **React** and **Vite** frontend against the API.

### Handwritten Digits Recognition | Python, NumPy

July 2025 – August 2025

- Built a **neural network from scratch** in **Python** and **NumPy**, implementing **feedforward propagation**, **backpropagation**, and **gradient descent** with no external ML frameworks.
- Trained on **MNIST** to over **95% accuracy**, tuning learning rate, hidden-layer size, and batch size by hand.

## TECHNICAL SKILLS

**Languages:** Python, Java, TypeScript, JavaScript, Swift, SQL (PostgreSQL, MySQL, SQLite), HTML, CSS

**Frameworks/Libraries:** React, Hono, Django, Flask, Node.js, TanStack Router, Tailwind CSS, Zod, Three.js, ARKit, NumPy, Matplotlib

**Developer Tools:** Git, GitHub Actions, Docker, Vitest, Supabase, Cloudflare Workers, AWS, pnpm, ESLint/Prettier, Vite, Neovim, Bash/Zsh